AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2023

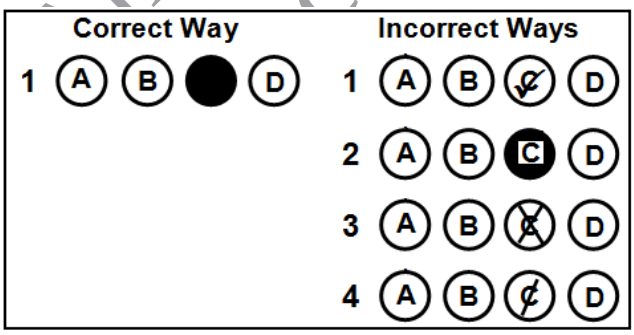
Computer Science Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. The type of operating system that is used in space research program is

   A. real-time operating system.
   B. multitasking operating system.
   C. time-sharing operating system.
   D. multiprogramming operating system.

2. Disk Operating System (DOS) is an example of

   A. time-sharing operating system.
   B. single-user operating system.
   C. multi-user operating system.
   D. real-time operating system.

3. Consider the given three processes **X**, **Y** and **Z** that are ready for execution. The operating system will manage the CPU time as follows:

   Process **X**: 3 milli sec
   Process **Y**: 1 milli sec
   Process **Z**: 4 milli sec

   The fastest executed sequence will be

   A. XYZ
   B. ZYX
   C. ZXY
   D. YXZ

4. All of the following are functions of operating system EXCEPT

   A. interrupt handling.
   B. process management.
   C. network management.
   D. compiling a computer program.

5. An operating system loads multiple programs, processes, tasks and threads in main memory and executes them at the same time. This function of operating system is called

   A. multiprogramming.
   B. multiprocessing.
   C. multithreading.
   D. multitasking.

6. The function of an operating system that allows to create user groups and assign privileges to them is

   A. I/ O management.
   B. user management.
   C. protection system.
   D. network management.

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7. Consider the given tasks.

I. Design the database
II. Design the flowchart and algorithms
III. Design the actual user interface screen
IV. Generate the code for dataflow process

The tasks performed in the development phase of System Development Life Cycle (SDLC) are

A. II and III.
B. II and IV.
C. I, III and IV.
D. II, III and IV.

8. The first phase of SDLC is

A. feasibility study.
B. planning phase.
C. defining phase.
D. analysis phase.

9. The symbol that indicates the end of a statement in a C++ program is

A.)
B. >
C. ;
D. }

10. Which of the following is/ are allowed in variable names?

I. Digits (0-9)
II. Underscore (_)
III. Blank space

A. II only
B. III only
C. I and II
D. I and III

11. The number of bytes an unsigned long int occupies is

A. 2
B. 4
C. 8
D. 10

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12. Consider the given C++ code.

```
short int var1, var2
double var3, var4
char var5
float var6
```

In the given C++ code, the total memory storage occupied by short int and float type variables is

A. 6 bytes.
B. 8 bytes.
C. 12 bytes.
D. 16 bytes.

13. The escape sequence that is used to move the cursor to the beginning of current line is

A. \b
B. \n
C. \r
D. \t

14. If a user wants to display all kinds of output on the screen, then the output function used will be

A. cin();
B. gets();
C. puts();
D. cout();

15. The highest value that can be stored in a variable having short int data type is

A. 32768
B. 32767
C. -32767
D. -32768

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Use the given C++ program to answer Q.16 and Q.17.

```
#include <iostream>
using namespace std;
int x, y, z;
int main ()
{
int i=2, j=4;
i==4;
j=7;
x=j-i;
y=i*j;
cout<<-x<<"\n";
cout<<y;
return 0;
}
```

16. The number of global variables present in the given program is

    A. 2
    B. 3
    C. 4
    D. 5

17. The output of this program is

5 14	5 -14
A	B
-5 14	-5 -14
C	D

18. Consider the given expression.

    (5%3)||(!(1)&&0)

    The number of binary operator(s) in this expression is

    A. one.
    B. two.
    C. three.
    D. four.

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19. Consider the given syntax.

```
if (condition)
{
        if (condition)
        {
                //statement
        }
        else
        {
                //statement
        }
        //statement
}
```

The structure identified from the given syntax is

A. if-else statement.
B. else-if statement.
C. nested if statement.
D. simple if statement.

20. The MOST suitable programming structure to determine the validity of a triangle is

A. loop.
B. array.
C. string.
D. selection.

21. Consider the given C++ program.

```
#include <iostream>
using namespace std;
int main()
{
        for (int i=15; i>1; i-=3)
        {
                cout<<i<<" ";
        }
}
```

The output of this program will be

A. 15 12 9 6 3 1
B. 15 12 9 6 3
C. 12 9 6 3 1
D. 12 9 6 3

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22. In C++ programming, multiple statement **for loop** is terminated with a

A. colon.
B. semicolon.
C. right brace.
D. right bracket.

Use the given C++ program to answer Q.23 and Q.24.

```
#include <iostream>
using namespace std;
int main()
{
   int num;
   cout<<"Enter an integer value"<<endl;
   cin>>num;
   if(num<=0)
   {
      if(num<0)
      {
      cout<<"Red"<<endl;
      }
      else
      {
      cout<<"Yellow"<<endl;
      }
   }
   else
   {
   cout<<"Orange"<<endl;
   }
return 0;
}
```

23. If the input of this program is 1, then the output will be

A. Red.
B. Orange.
C. Yellow.
D. Syntax error.

24. To get the output Yellow, the input should be

A. -2
B. -1
C. 0
D. 1

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25. All of the following objectives can be achieved using loop statement EXCEPT

A. print first 10 numbers of Fibonacci series.
B. print the sum of first 5 odd numbers.
C. determine the validity of a triangle.
D. calculate the factorial of a number.

26. Consider the given C++ program.

```
#include <iostream>
using namespace std;
int main ()
{
   int j=20;
   do
   {
      j-=3;
      cout<<++j<<endl;
   } while(j<14 );
return 0;
}
```

Upon execution, the program will display

A. 20
B. 18
C. 17
D. 14

27. An array is declared by specifying its

A. data type followed by its size and then its name.
B. data type followed by its name and then its size.
C. name followed by its data type and then its size.
D. name followed by its size and then its data type.

28. The size of array is specified inside

A. round brackets.
B. square brackets.
C. curly brackets.
D. angle brackets.

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29. When values from an array are read using a loop, what effect (change or remain constant) will it have on the aspects of array element (index number, name and value)?

	Index Number of Array Element	**Name of Array Element**	**Value of Array Element**
A	Remains Constant	Changes	Remains Constant
B	Remains Constant	Changes	Changes
C	Changes	Remains Constant	Remains Constant
D	Changes	Remains Constant	Changes

30. Sana has written a C++ program. To perform a specific task, she has written a piece of code in the program in 5 different places.

A programming concept that she could have used to write the code once and use it in as many places as she wants is

A. loop.
B. array.
C. pointer.
D. function.

31. Consider the given list of functions.

I. div()
II. add()
III. sub()
IV. abs()

The built-in functions in this list are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

32. A return type that a function may have is

A. loop.
B. array.
C. pointer.
D. pre-processor.

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33. The statement that shows the CORRECT use of parameters with a user-defined function named as **ADD** is

int ADD(int x, int y)	int ADD(x, y);
A	B
ADD(x, y);	ADD(x, y)
C	D

Use the given C++ program to answer Q.34, Q.35 and Q.36.

```
#include <iostream>
using namespace std;
void G(int x, int y, int z)
{
   x = x + y;
   y = z + x;
   z = x + y;
   cout << x << endl;
   cout << z - y << endl;
   cout << x + z;
}
int main()
{
   int a = 2, b = 3, c = 4;
   G(a,b,c);
   return 0;
}
```

34. A component of user defined function that is NOT present in this program is

A. function header.
B. calling the function.
C. function declaration.
D. invoking the function.

35. The number of actual parameters in this program is

A. two.
B. three.
C. five.
D. six.

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36. The output of the given program is

5 5 19	2 1 6
A	B
5 9 14	2 3 4
C	D

37. Consider the given C++ program.

```
#include <iostream>
using namespace std;
int main () {
  int a = 9.9;
  int  *b;
  b = &a;
  cout << "a variable: " << a << endl;
  cout << "b variable: " << b << endl;
  cout << "*b variable: " << *b << endl;
  return 0;
}
```

The output of this program will be

(**Note**: The memory address in the output of this program may vary upon execution.)

a variable: 9 b variable: 0x6ffe04 *b variable: 9	a variable: 9 b variable: 0x6ffe04 *b variable: 10
A	B
a variable: 9.9 b variable: 0x6ffe04 *b variable: 9.9	a variable: 9.9 b variable: 0x6ffe04 *b variable: 10
C	D

38. The memory address of a variable is stored in a/ an

A. object.
B. pointer.
C. function.
D. constructor.

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39. The CORRECT way to declare a pointer variable is

A. int x*;
B. int *x;
C. int &x;
D. int x&;

40. All of the following are important pillars of object-oriented programming (OOP) EXCEPT

A. abstraction.
B. typecasting.
C. encapsulation.
D. polymorphism.

41. The process of hiding irrelevant details from the user is called

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

42. Operator overloading and function overloading are examples of

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

43. Which of the following is FALSE about constructors?

A. It has integer return type.
B. It initialises objects of a class.
C. It can take one or more arguments.
D. Its name is the same as class name.

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Use the given C++ program to answer Q.44 and Q.45.

```
#include <iostream>
using namespace std;
class SampleClass
{
  public:
    SampleClass()
{
      cout << "Hello World!";
 }
};
int main()
{
  int A1, A2;
  SampleClass X1;
  SampleClass X2;
  return 0;
}
```

44. The number of access specifiers used in this program is

    A. one.
    B. two.
    C. three.
    D. four.

45. The number of objects declared in this program is

    A. one.
    B. two.
    C. three.
    D. four.

46. An object **X** of class **Y** will be declared as

    A. X Y;
    B. Y X;
    C. X, Y;
    D. Y, X;

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47. The data member and member functions can be accessed ONLY in the same class.

With reference to the given statement, the option that shows the access specifiers used to give this access to these members is

	Access Specifiers		
	Public	**Protected**	**Private**
A	yes	yes	yes
B	yes	yes	no
C	no	no	yes
D	no	yes	yes

48. Which of the following classes are related to C++ file handling with respect to output operations?

I. ofstream
II. ifstream
III. fstream

A. I only
B. II only
C. I and III
D. II and III

49. A command of C++ language used to open a file named as **mydata.dat** file in the binary mode is

ofstream myFile; newFile.open("mydata.dat", ios::binary);	ofstream newFile; myFile.open("mydata.dat", ios::binary);
A	B
ofstream myFile; myFile.open("mydata.dat", ios::bin);	ofstream myFile; myFile.open("mydata.dat", ios::binary);
C	D

50. A sample file named as abcfile.txt has the following data.

Day 10

To read data from this file, the number of character arrays required is

A. 1
B. 2
C. 5
D. 6

END OF PAPER

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